Unit 7 Pre-Test

Quadratic Functions

Standards assessed: NC.M1.F-IF.4 · F-IF.7 · F-IF.9 · A-CED.1 · A-CED.2 · F-BF.1b

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.F-IF.4 · NC.M1.F-IF.7 Read the key features of a parabola.

• *I can tell which way a parabola opens and find its vertex, axis of symmetry, y-intercept, and maximum or minimum.*

Answer each question. Use the picture to help you.

How to type your answers

point `(2, 5)` line `x=-1` expression `x^2+2x-2` inequality `y<=6`

1. *Direction*

Does $y = -2x^2 + 3$ open up or down?

$-2x^2$ $+3$

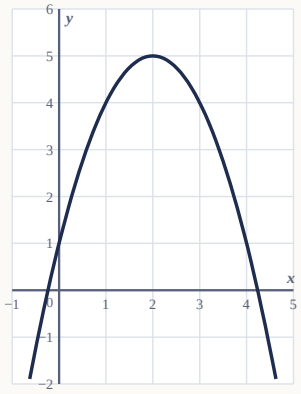
$a = ?$

Answer: _____

`answer:` down (a = −2 is negative)

2. *Vertex*

What is the vertex of the parabola?

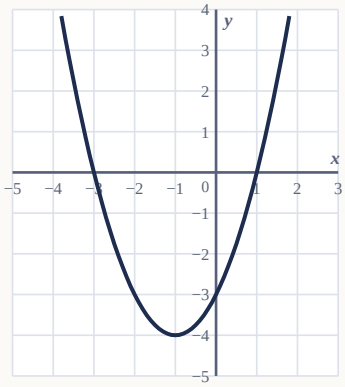


Answer: _____

`answer:` (2, 5)

3. *Axis of symmetry*

What is the axis of symmetry of the parabola?

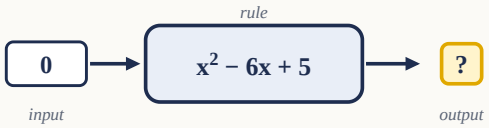


x = _____

answer: x = −1

4. *y-intercept*

Find the y-intercept of $f(x) = x^2 - 6x + 5$.



Answer: _____

answer: 5 (the point (0, 5))

5. *Table of values*

$f(x) = x^2 + 2x - 1$. Find f(3).

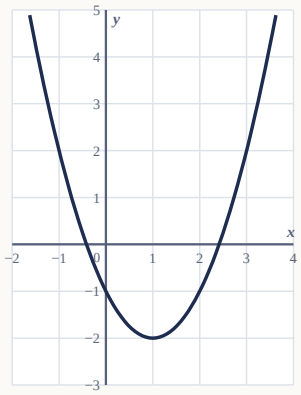
x	1	2	3
f(x)	2	7	?

f(3) = _____

answer: 14

6. *Max or min?*

Is the vertex of this parabola a maximum or a minimum?



Answer: _____

answer: minimum

SCORE 4.0 Target Content

NC.M1.F-IF.4 · NC.M1.F-IF.7 · NC.M1.F-IF.9 · NC.M1.A-CED.2 · NC.M1.F-BF.1b Use forms of quadratics, compare functions, and combine functions.

• *I can find the vertex from standard or vertex form, give the range, compare two quadratics, write an equation, and add functions.*

Solve each problem. Show your work on paper.

7. *Find the vertex*

Find the vertex of $y = x^2 - 6x + 8$.

a = 1, b = −6, c = 8

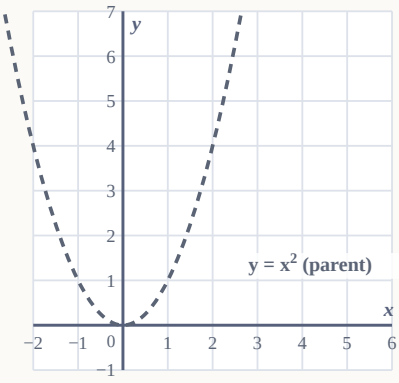
$$x = \frac{-b}{2a} = ?$$

Answer:

answer: (3, −1)

8. *Vertex form*

$y = (x - 4)^2 + 1$ is the parent graph $y = x^2$ moved. What is its vertex?

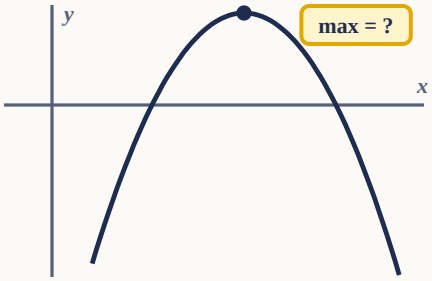


Answer:

answer: (4, 1)

9. *Range*

What is the range of $y = -(x - 1)^2 + 6$?

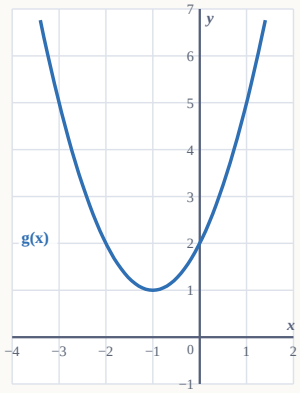


Answer:

answer: $y \le 6$

10. *Compare*

$f(x) = x^2 - 4x + 7$. The function g is graphed. Which function has the smaller minimum value: f or g?

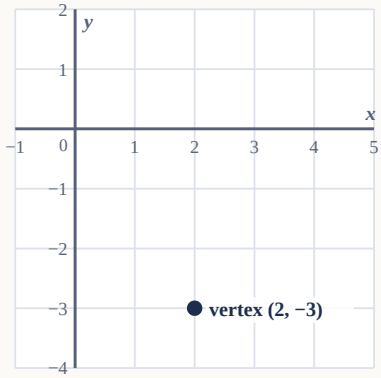


Answer:

answer: g (min 1 < 3)

11. *Write the equation*

Write the equation of the parabola with vertex (2, −3) and a = 1. Use vertex form.



y = _____

answer: $y = (x - 2)^2 - 3$

12. *Add functions*

$f(x) = x^2 - 3$ and $g(x) = 2x + 1$. Write $(f + g)(x)$.

x	0	1	2
f(x)	−3	−2	1
g(x)	1	3	5
(f + g)(x)	?	?	?

(f + g)(x) = _____

answer: $x^2 + 2x - 2$

SCORE 5.0 Advanced Content

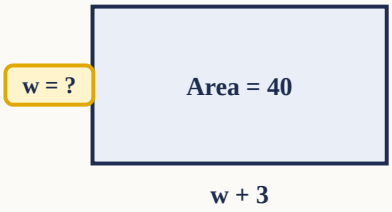
NC.M1.F-IF.4 · NC.M1.F-IF.9 · NC.M1.A-CED.1 · NC.M1.A-CED.2 · NC.M1.F-BF.1b Model with quadratics and build new functions.

• *I can solve area and height problems, multiply functions, compare maximums, and write a quadratic from its zeros and a point.*

Solve each problem. Some answers are words.

13. *Area equation*

A rectangle's length is 3 more than its width. Its area is 40. Find the width.

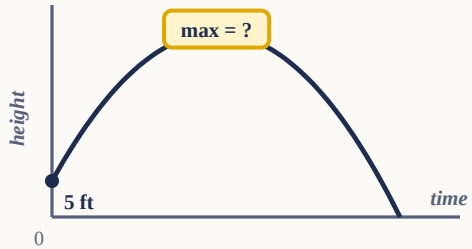


w = _____

answer: 5

14. *Maximum*

$h(t) = -16t^2 + 64t + 5$ is a ball's height in feet. What is its **maximum** height?



feet = _____

answer: 69 feet

15. *Multiply functions*

$L(x) = x + 5$ and $W(x) = x - 2$ are a rectangle's length and width. Write $A(x) = L(x) \cdot W(x)$ in standard form.

	x	+5
x	?	?
−2	?	?

A(x) =

answer: $x^2 + 3x - 10$

16. *Compare*

$f(x) = -x^2 + 4x$. The table shows g. Which has the greater maximum: f or g?

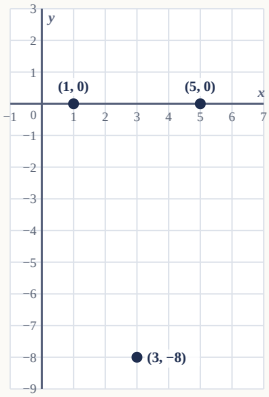
x	0	1	2	3	4
g(x)	1	4	5	4	1

Answer:

answer: g (max 5 > 4)

17. *Write the equation*

A parabola has zeros 1 and 5 and passes through (3, −8). Write its equation in standard form.

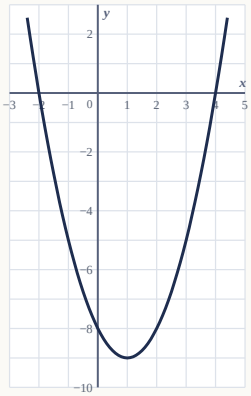


y =

answer: $y = 2x^2 - 12x + 10$

18. *Decreasing*

$y = x^2 - 2x - 8$. For which x-values is the function **decreasing**?



Answer:

answer: $x < 1$